

Vatsal Jha

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Bangalore, India

Research engineer building and evaluating ML systems for multi-agent autonomy, federated learning, and decision-making under uncertainty across research and production environments.

Selected Experience

testAIing

Jan–Jul 2026

Machine Learning Intern

Remote

- Owned ML R&D (detecting AI-generated/deepfake speech: ECAPA-TDNN, WavLM, mixture-of-experts; **0.94–0.96 AUC**, **1.97% EER** after Hindi fine-tuning) and the production DevOps migration (monolith→Kubernetes, HPA, Nginx, observability) for the same service. **1st Prize/70+ interns (research presentation competition)**. [blog]

Samsung Research India

May–Jul 2025

Software Engineering Intern, Server Platform / MAGPIE

Bangalore

- Individually designed and built a multi-agent RCA system**, spanning literature review, implementation, evaluation and handoff. *Tooling*: Google ADK/MCP interfaces over Prometheus, Kubernetes, Jaeger, OpenSearch and Milvus. *Reasoning*: evidence/hypothesis validation, cross-incident memory and diagnosis-only guardrails.
- Evaluation*: built an OpenTelemetry + Chaos Mesh testbed against a published **90-scenario** microservice fault-injection benchmark (9 fault types × 10 incidents), with strong Top-1 root-cause identification on an internal dev slice; received a full-time **pre-placement offer (PPO)**.

Samsung R&D Institute Bangalore – PRISM

Oct 2024–May 2025

Team Lead / Co-primary Researcher

Hybrid

- Co-led 4-student **HOMA** team: multilingual SLM orchestration decomposing multi-device commands into sequential/concurrent execution across 11+ languages; synthetic evaluation + Gemma/Qwen/Phi-family edge benchmarks. **Samsung PRISM Excellent Worklet Award**; presented at ICCCNT 2025. [code]

Council on Energy, Environment and Water (CEEW)

Jun–Aug 2024

Data Science Intern

Delhi

- Built a BERTopic/transformer semantic-matching pipeline over **93K task/skill segments**, **600+ occupations** and a public job-transition prototype; this early prototype later informed the public **CEEW–ADB SMART** platform and a 2026 CEEW coal-workforce report. [HF] [SMART] [report]
- Co-developed Delhi PM2.5 forecast bias correction (MAPE **43%→27%**), IICAQM 2024 poster; built a solar/BESS decision tool. [solar]

NTNU × BITS Pilani

Mar 2024–Aug 2025

Remote Research Collaborator, with Dr. Sukalpa Chanda

Remote

- Built an end-to-end handwriting detection+recognition pipeline on IAM: shared ResNet-50 backbone, FOTS-style region detector, rotated ROI alignment, attention-augmented PHOSC word recognizer, jointly trained under constrained compute; **6–8% CER** on selected internal development/test runs.

VIT × King Saud University

Dec 2022–Jul 2024

ML Research Intern, advised by Dr. B. Ashok

Hybrid

- Designed the stochastic trip-segmentation ML pipeline for EV SoC prediction from CAN/BMS telemetry, with King Saud University (Faisal Alsaif) under a Royal Academy of Engineering grant; **test MSE $\approx$ 0.041**; published *Journal of Energy Storage* (IF $\approx$ 11, Q1) 97 (2024); Raman Research Award. Continuing under Dr. Ashok (Aug 2024–Present) on federated recommendation, EV-charging placement, and carbon reporting (see Manuscripts under review). [paper] [code]

Education

Vellore Institute of Technology

2022–2026

B.Tech. Computer Science & Engineering (IoT)

Vellore, India

CGPA **9.23/10**; **Top 1%** nationally, NPTEL RL (IITM) – invited to interview for RL Instructor, IITM BS; **Top 5%** Responsible & Safe AI (IIITH × IITM). **Final Year Project**: SOPRA (see Selected Research).

Selected Research

IEEE ICDSINC 2025, Best Paper: CA-AMFA. First-author, with Dr. Yokesh Babu S; Raspberry Pi adaptive MFA, bandit-inspired weighting; Thompson Sampling cut simulated FAR **44.75%**. [paper] [code]

Open-source: Maieutic Research. Socratic Claude Code skill gating autonomy via predict-then-observe logs. [code]

PromptTrend. Contextual prompt recommendation API. [code] [PyPI] **BANDITS-IDS**. Bandit-based IDS. [code]

Manuscripts under review:

- SOPRA**, LLM Java test repair supervised by Dr. Sathya K: **58.7% EM**, **+14.7 pp** vs baseline. *SANER 2027*.
- HERO** treats each full charging-station layout as a bandit arm for SUMO-in-the-loop placement search; **+39.9%** composite reward vs. best baseline across 13 methods. *Elsevier EAAI*.
- BehaviorFL/BehaviorFL-MT**, privacy-preserving federated in-vehicle recommendation under heterogeneous sensors & partial supervision, co-inventor (Published Patent App. **202641089937**); discovery-aware sampling gives **5.8×** rare-task exposure at fixed budget. *NeurIPS 2026 NewInML Workshop*.
- Ranking Is Not Valuation** audits carbon-claim overstatement from post-selection bias; reporting bound cuts overstatement **51%→11%** without changing the recommended product; patent in preparation. *NeurIPS 2026 climate workshop*.

Selected Recognition & Skills

Recognition: Samsung PRISM Excellent Worklet Award; 2x National Hackathon Winner (Samsung PRISM [code]; IISF 2023 Space Hackathon [code]); Top-3 KSP Datathon [code].

Leadership: *Co-Founder, Team Prometheus* (30+ members) [site]; led zero-to-one RoboCup SSL AI R&D, spearheaded RL behavior prototyping, secured major in-kind engineering sponsorships (SolidWorks, ANSYS, Altium), and mentored the team behind the HINT'25 NMPC paper [code]. *Vice Chair→Chair, Youth Red Cross VIT* (750+ donations).

ML/Research: PyTorch, TensorFlow, HF, vLLM, LLMs/agents, RAG, RL/RLHF, FL. **Systems**: Docker, Kubernetes, FastAPI, MCP/ADK, Redis/PostgreSQL/Milvus, Prometheus/Grafana. **Languages**: Python, Java, C++.